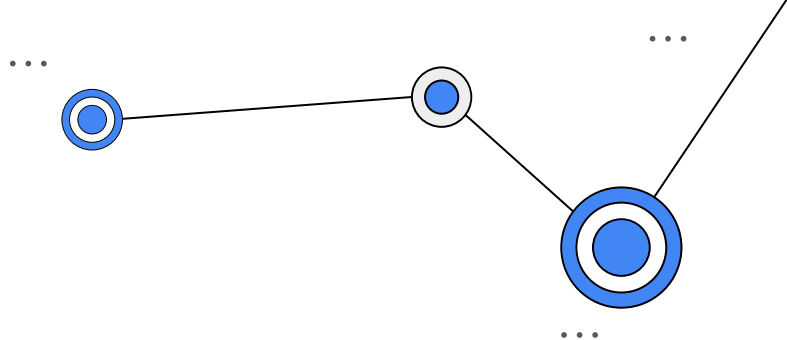




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DATA-DRIVEN SYSTEMS ENGINEERING

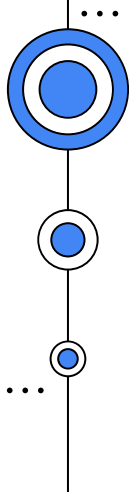
L15 - MLOps: Clustering Analysis

Matheus Camilo da Silva
matheus.camilodasilva@phd.units.it

15

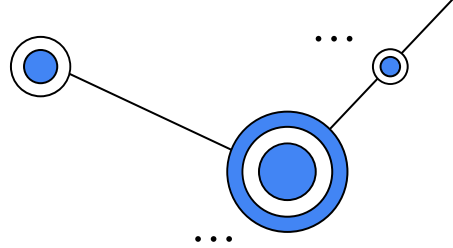
Clustering Analysis

L15 - MLOps: Clustering Analysis



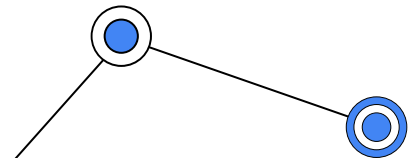
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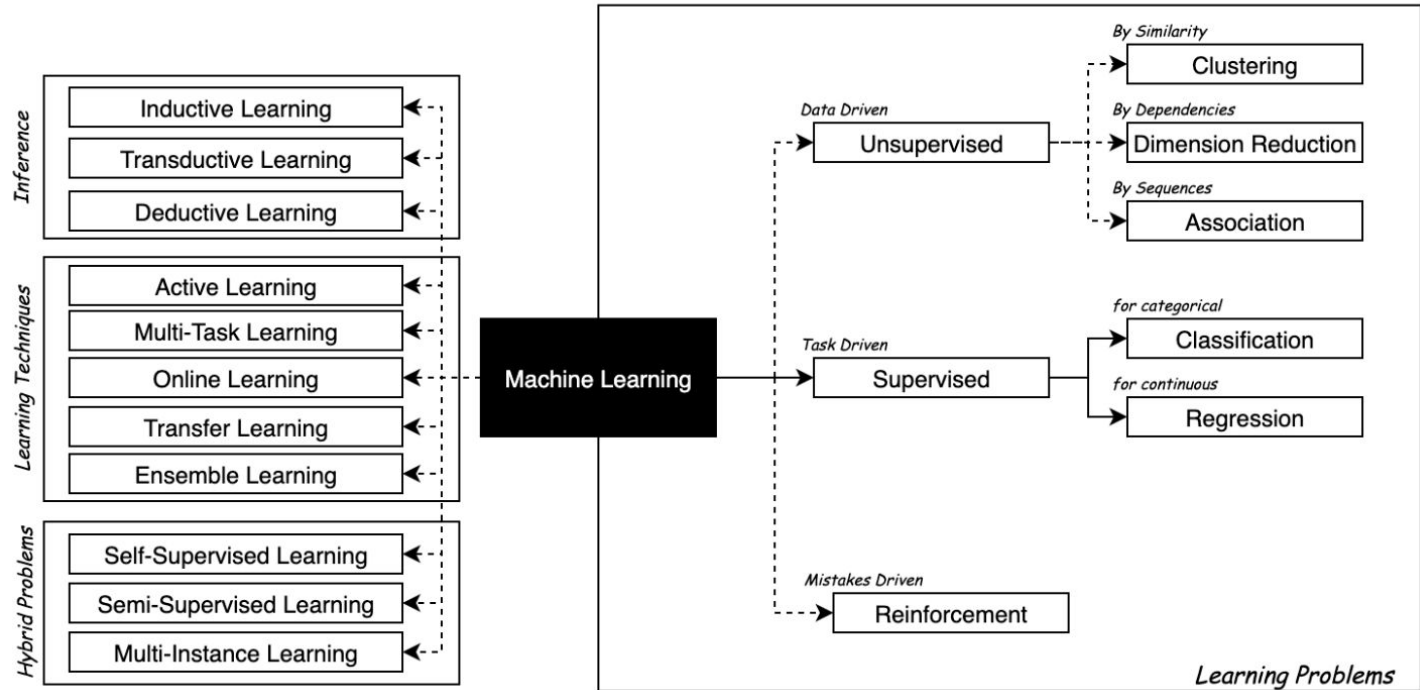


Our Agenda:

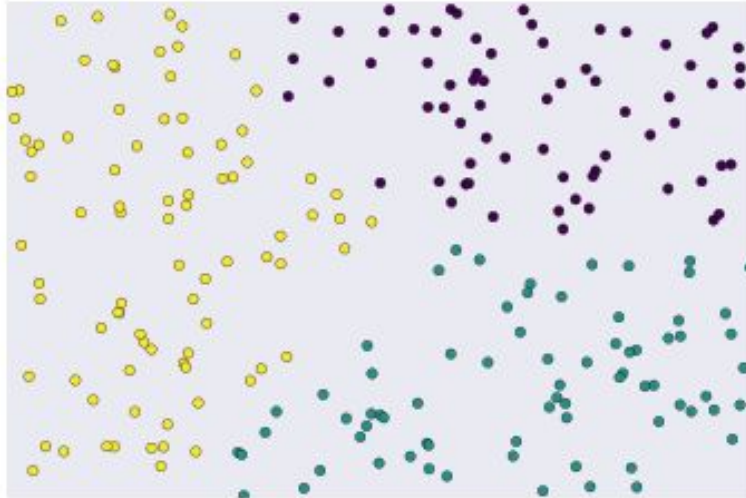
1. Machine Learning Tasks.
2. What is Clustering?
3. What can you do with Clustering?
4. Clustering Example.
5. Similarity and Dissimilarity Measures
6. How to Choose the Right Measure?
7. Clustering algorithms.
8. Challenges and Considerations in Clustering for MLOps.
9. Unsupervised ML, beyond clustering.
10. Principal Component Analysis ?



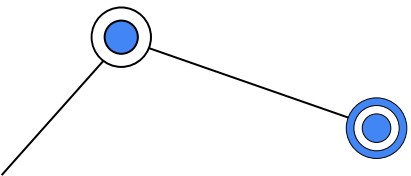
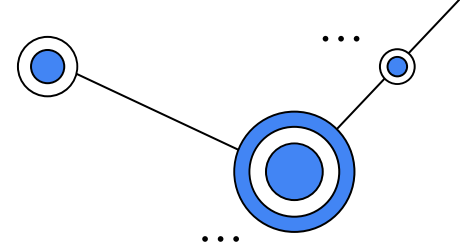
Machine Learning Tasks



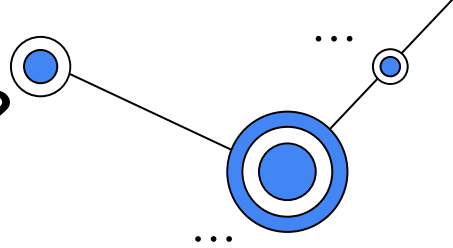
What is **Clustering**?



Unlabeled examples grouped into three clusters.



What can you do with Clustering?

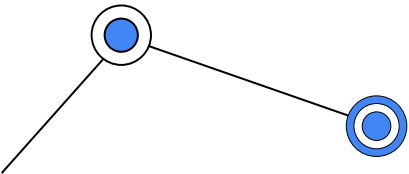


Generalization

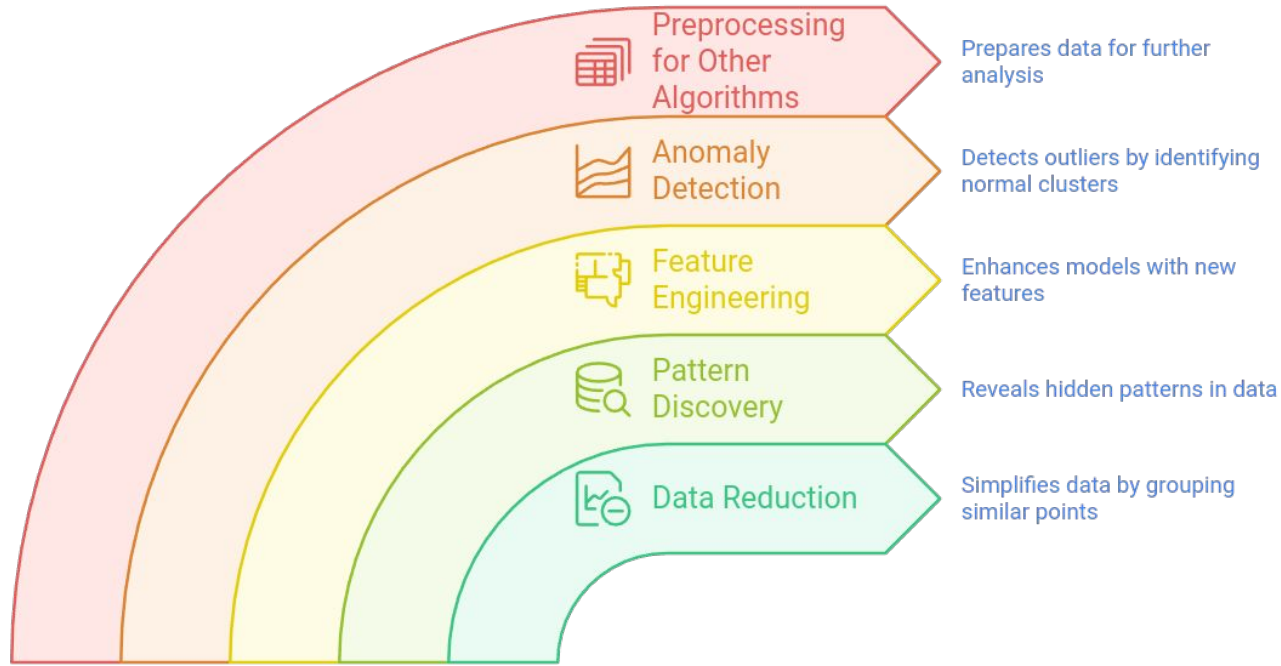
When some examples in a cluster have missing feature data, you can infer the missing data from other examples in the cluster.

Data Compression

As discussed, feature data for all examples in a cluster can be replaced by the relevant cluster ID. This replacement simplifies the feature data and saves storage.



What can you do with Clustering?

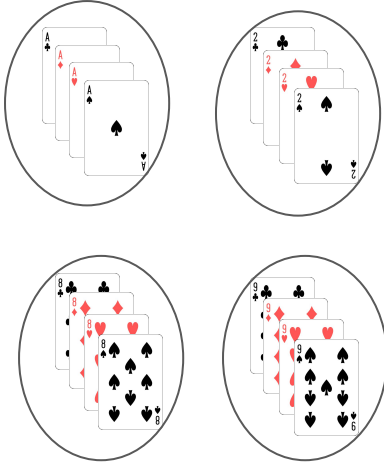


Problem

- Given a deck of cards, provide a clustering

Problem

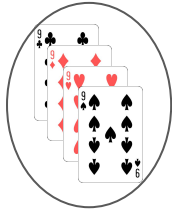
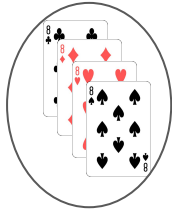
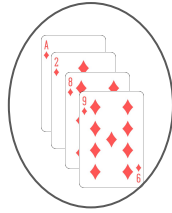
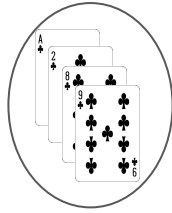
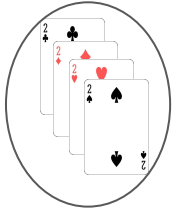
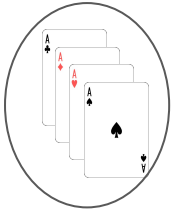
- Given a deck of cards, provide a clustering



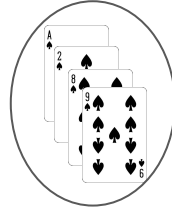
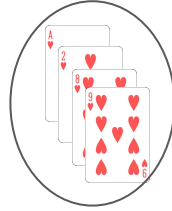
By numbers

Problem

- Given a deck of cards, provide a clustering



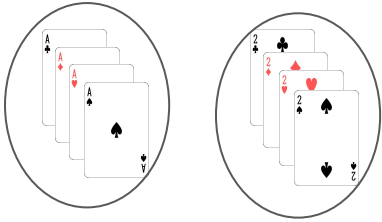
By numbers



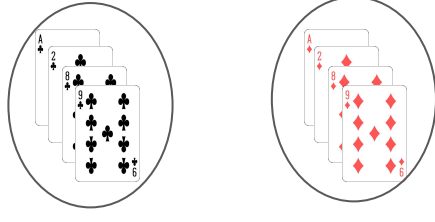
By suits

Problem

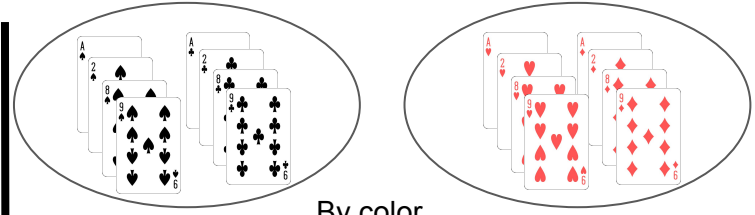
- Given a deck of cards, provide a clustering



By numbers



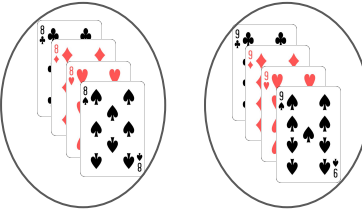
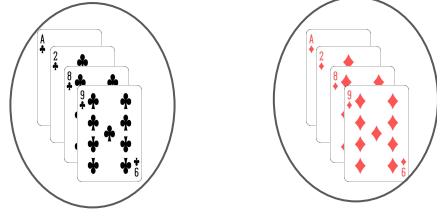
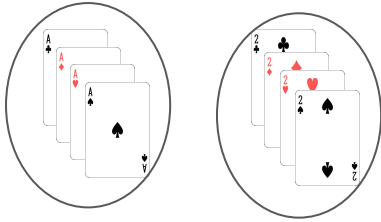
By suits



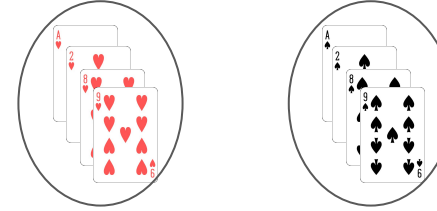
By color

Problem

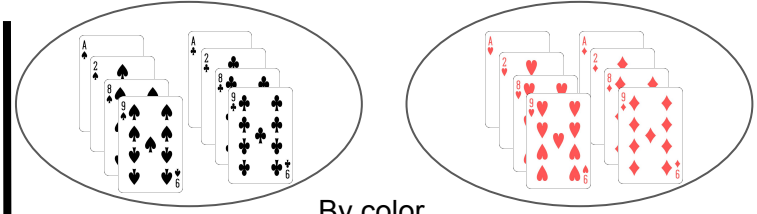
- Given a deck of cards, provide a clustering



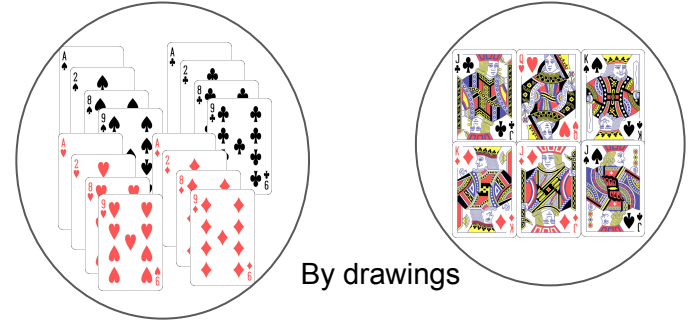
By numbers



By suits



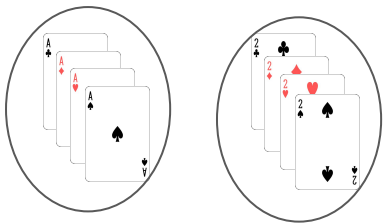
By color



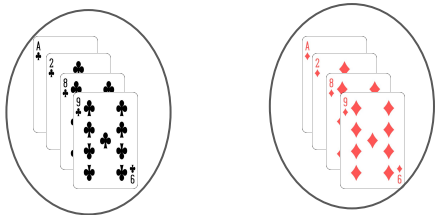
By drawings

Problem

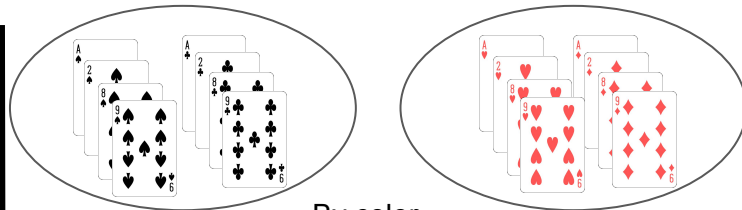
- Given a deck of cards, provide a clustering



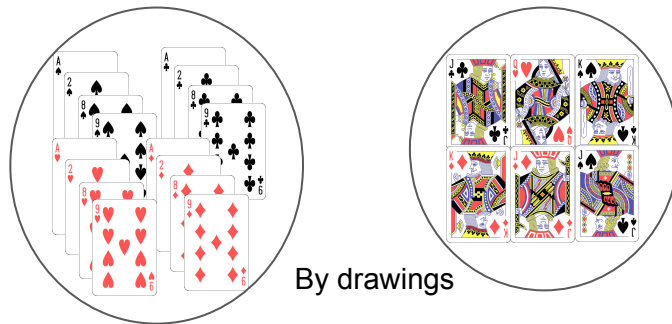
By numbers



By suits



By color



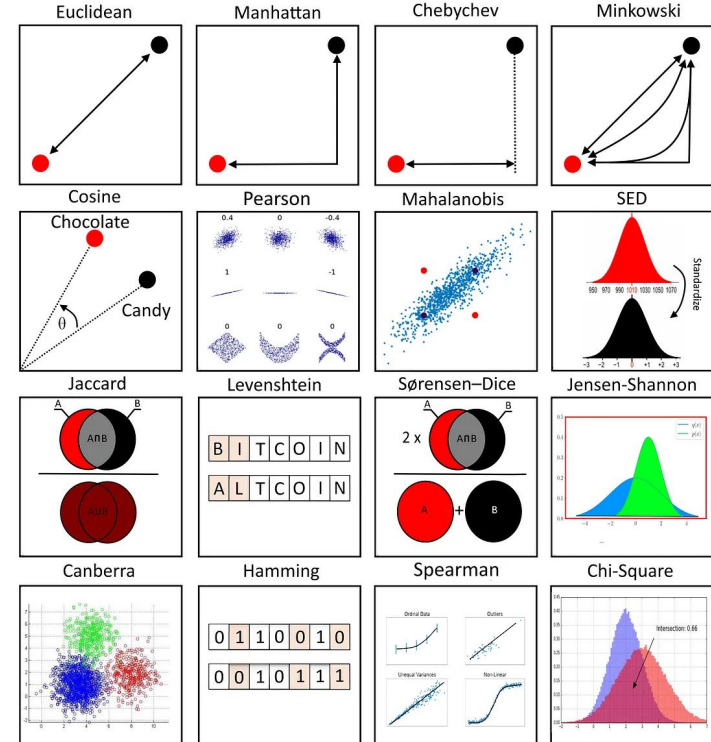
By drawings

- What are the *natural* structures?
- Clustering is not generalizable, i.e., it inherently needs to be *problem-centric*

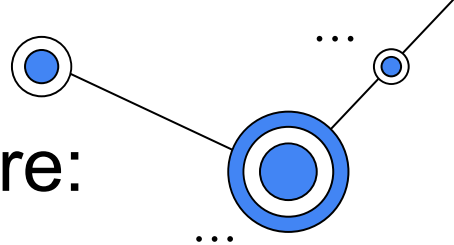
Similarity and Dissimilarity Measures

These measures help quantify the degree of resemblance or difference between data points or objects in a dataset.

They play a crucial role in determining which data points should be grouped together in clustering or how similar items are in recommendation systems.

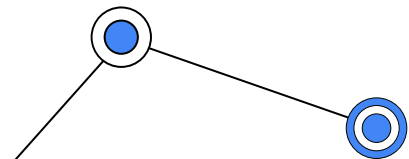


How to Choose the Right Measure:

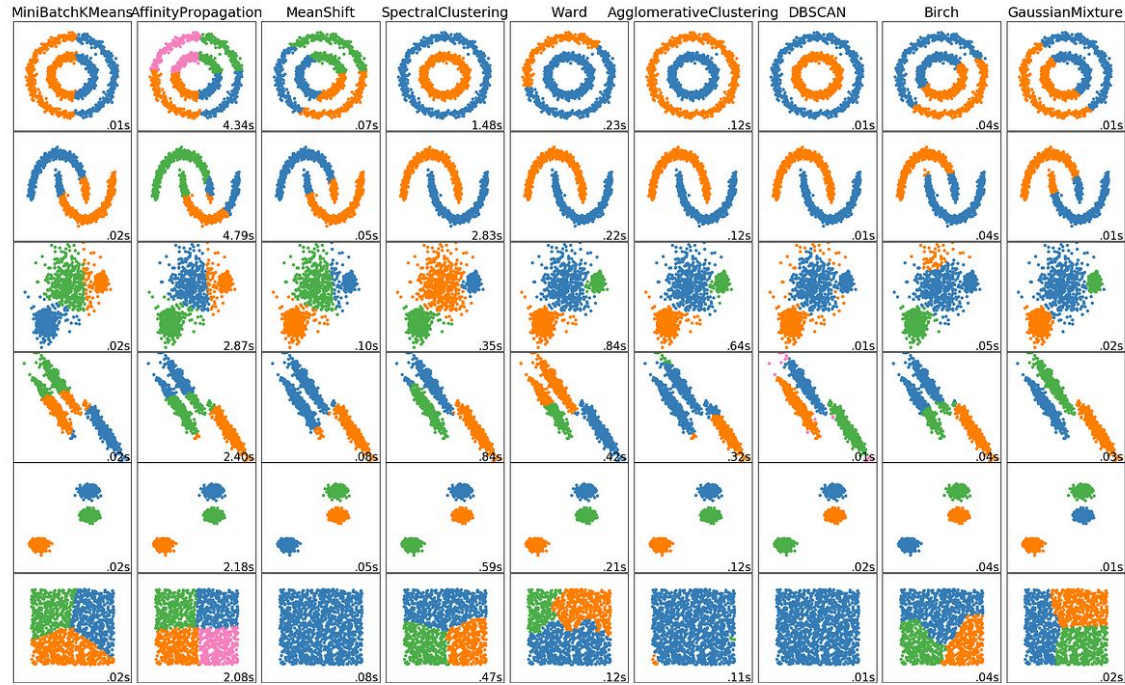


- The choice of similarity or dissimilarity measure depends on the nature of the data and the problem you are trying to solve.
- Continuous data often works well with **Euclidean distance or cosine similarity**.
- Categorical or binary data may require specialized measures like **Jaccard similarity/dissimilarity**.
- Data that is **centered** and **scaled**.

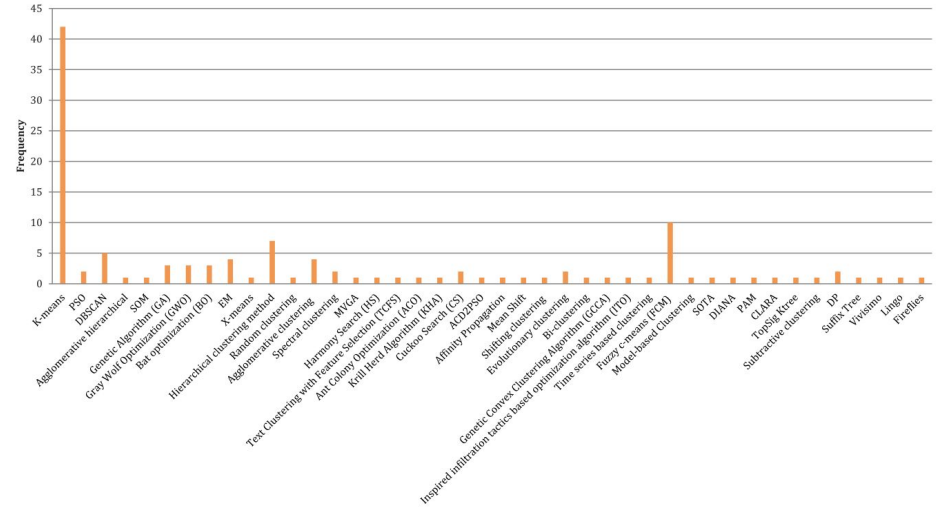
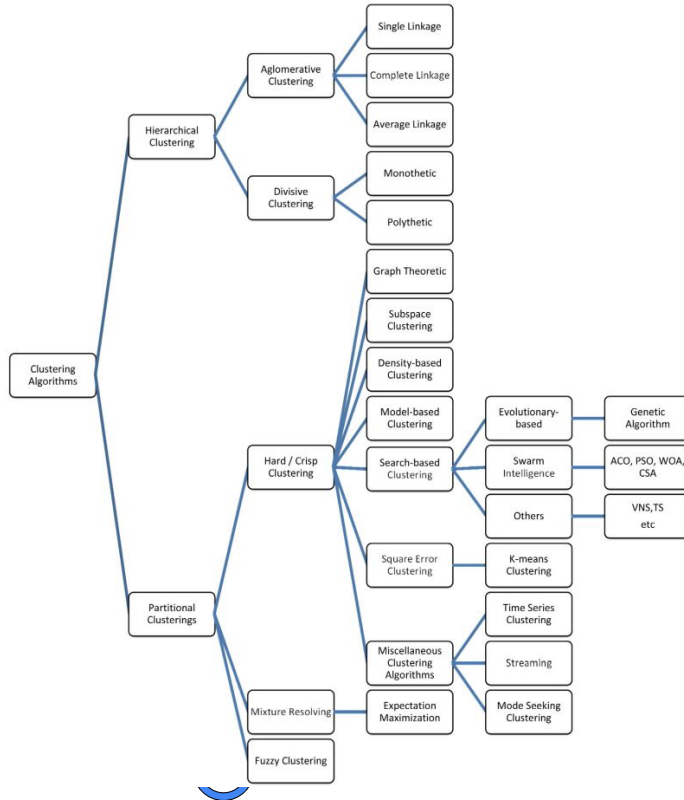
In practice, it's often a good idea to experiment with **different measures** and assess the quality of clustering or recommendations they produce, as the most appropriate measure can vary depending on the **specific dataset and application**.



Clustering Algorithms



Clustering Algorithms



Survey paper

A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects

Absalom E. Ezugwu^{a,*}, Abiodun M. Ikotun^a, Olaide O. Oyelade^a, Laith Abualigah^{b,c,d,e}, Jeffery O. Agushaka^a, Christopher I. Eke^a, Andronicus A. Akinyelu^a

^a School of Computer Science, University of KwaZulu-Natal, Pietermaritzburg Campus, Pietermaritzburg 3201, South Africa

^b Faculty of Computer Sciences and Informatics, Amman Arab University, Amman 11953, Jordan

^c School of Computer Sciences, Universiti Sains Malaysia, Pulau Pinang, 11800, Malaysia

^d Department of Computer Science, Faculty of Natural Science, Federal University of Lafia, Lafia, Nigeria

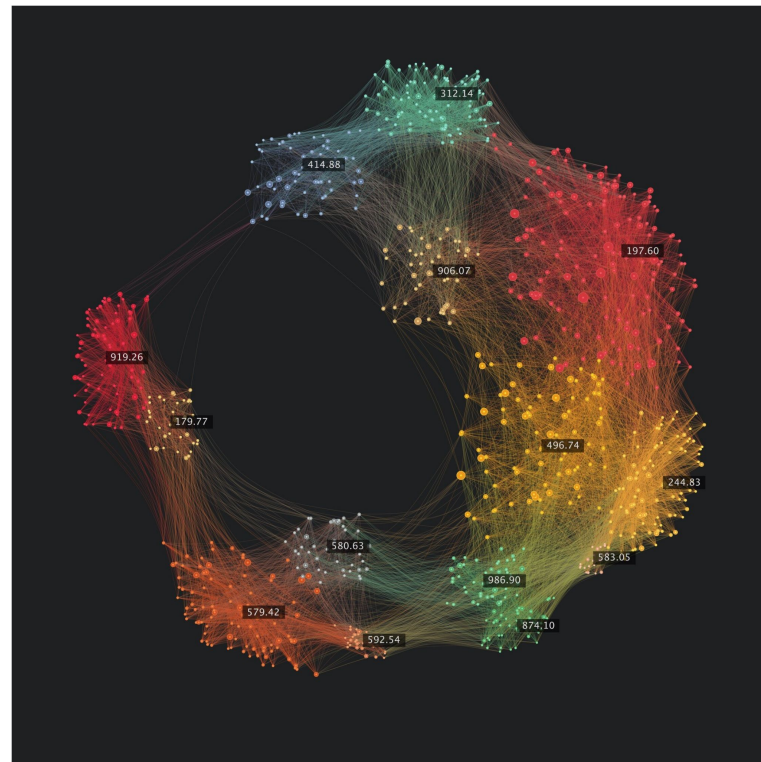
^e Department of Computer Science and Informatics, University of the Free State, 3301 Bloemfontein, South Africa



Clustering Algorithms

When choosing a [clustering algorithm](#), you should consider whether the algorithm **scales** to your dataset. Datasets in machine learning can have millions of examples, but not all clustering algorithms scale efficiently. **Asymptotic complexity:**

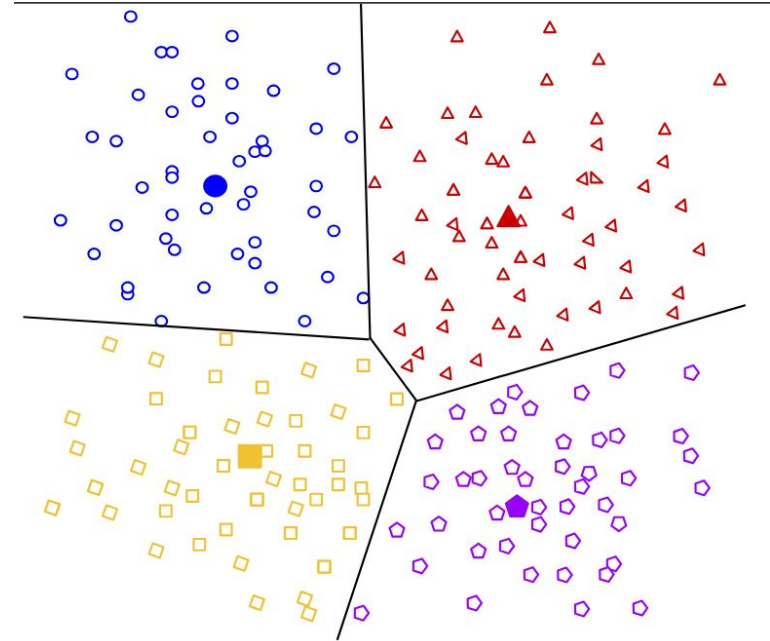
1. BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies): $O(n)$
2. DBSCAN (Density-Based Spatial Clustering of Applications with Noise): $O(n * \log(n))$
3. OPTICS (Ordering Points To Identify the Clustering Structure): $O(n^2 * \log(n))$
4. Agglomerative Clustering: $O(n^2 * \log(n))$
5. Spectral Clustering: $O(n^2 * \log(n))$
6. Mean Shift: $O(n^2)$
7. K-Means Clustering: $O(n * k * I * d)$, where n is the number of data points, k is the number of clusters, I is the number of iterations, and d is the number of features.
8. Hierarchical Clustering: $O(n^2 * \log(n))$



Centroid-based Clustering

Centroid-based clustering organizes the data into non-hierarchical clusters. **k-means** is the most widely-used centroid-based clustering algorithm.

Centroid-based algorithms are efficient but sensitive to initial conditions and outliers.

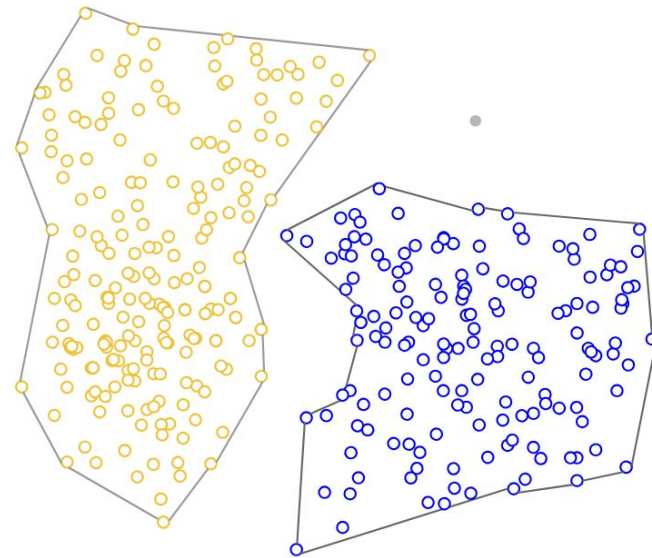


Example of K-mean Clustering

Density-based Clustering

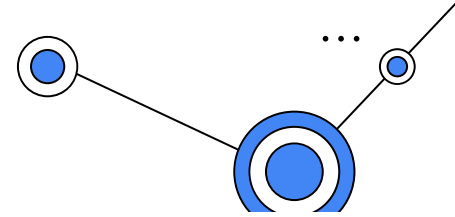
Density-based clustering connects areas of high density into clusters. This allows for **arbitrary-shaped distributions** as long as dense areas can be connected.

These algorithms have difficulty with data of varying densities and high dimensions. Further, by design, these algorithms do not assign outliers to clusters. Example: **DBSCAN**

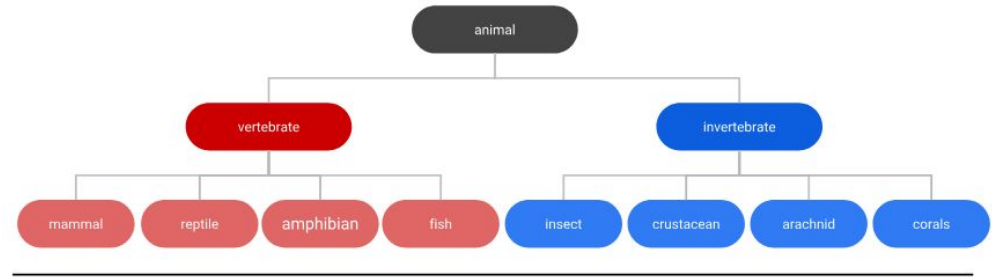


Example of DBSCAN

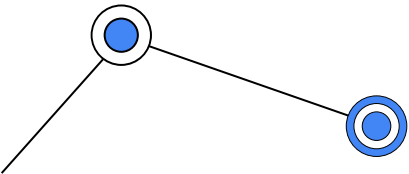
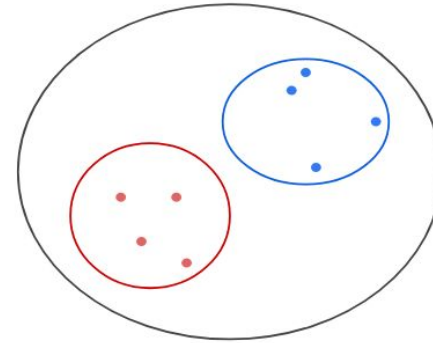
Hierarchical Clustering



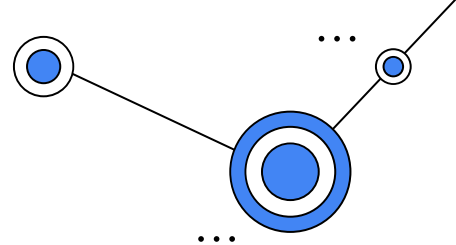
Hierarchical clustering creates a tree of clusters is well suited to **hierarchical data**, such as taxonomies. Example: Image Segmentation and Market Segmentation.



The traditional Hierarchical Clustering is based on ward's method, in which minimizes the variance within clusters. It is less sensitive to noise and outliers compared to the above methods and tends to create well-balanced clusters.



Clustering Validation Indexes



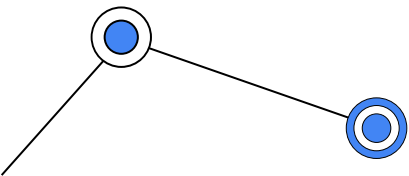
Internal Validation Metrics:

- ✧ **Silhouette Score:** Measures the quality of each data point cluster assignment by considering the distance between data points within the same cluster and the distance to the nearest neighboring cluster. A higher silhouette score indicates better cluster quality.
- ✧ **Davies-Bouldin Index:** Evaluates the average similarity between each cluster and its most similar cluster. Lower values indicate better cluster separation.
- ✧ **Dunn Index:** Measures the ratio of the minimum inter-cluster distance to the maximum intra-cluster distance. A higher Dunn index indicates better clustering.

External Validation Metrics:

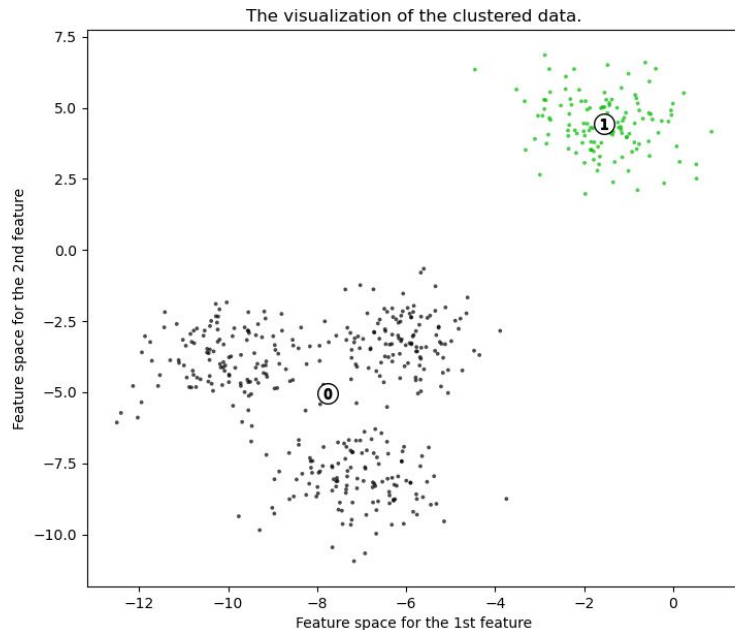
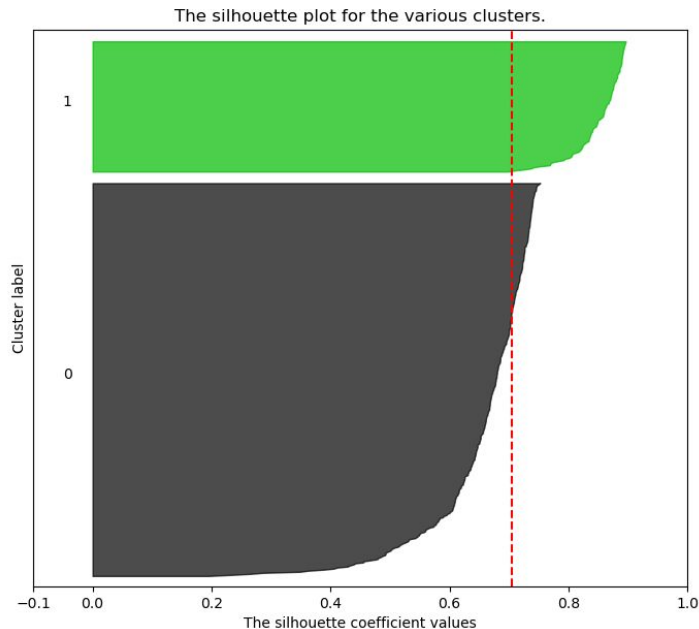
- ✧ **Adjusted Rand Index (ARI):** Measures the similarity between the true labels of data points and the cluster assignments. A higher ARI indicates better clustering.
- ✧ **Normalized Mutual Information (NMI):** Measures the amount of information shared between true labels and cluster assignments. Higher NMI values represent better clustering.

Visual Inspection: Visual inspection involves creating visualizations of the clustering results, such as scatterplots or dendrograms, to qualitatively assess the quality of clusters. This is particularly useful for understanding the structure of the data.

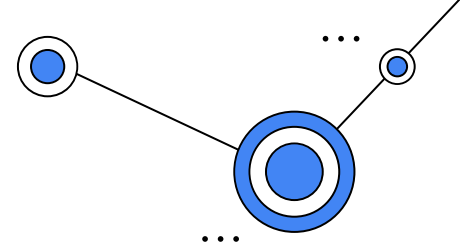


Silhouette Analysis

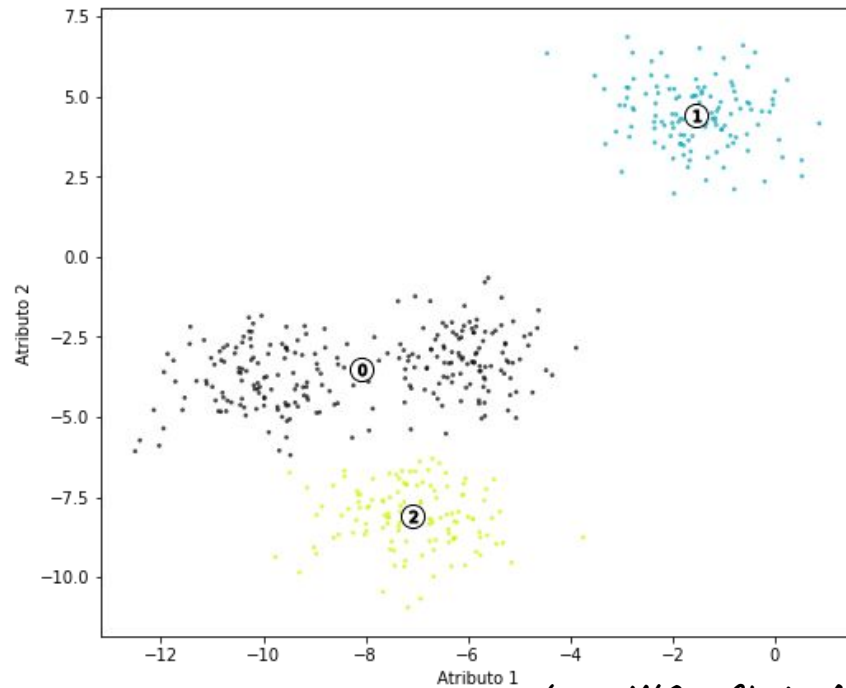
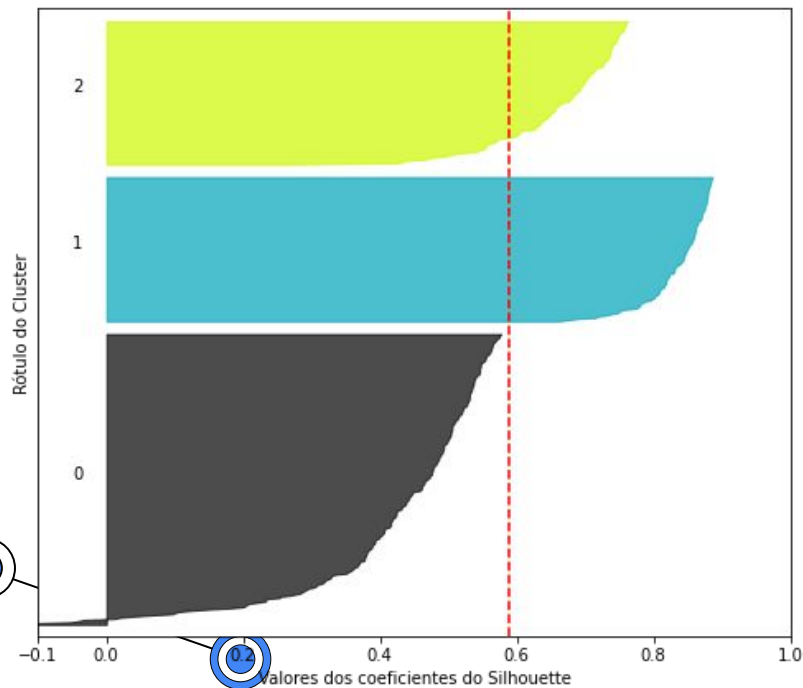
$k = 2$ (Silhouette avg : 0.79)



Silhouette Analysis

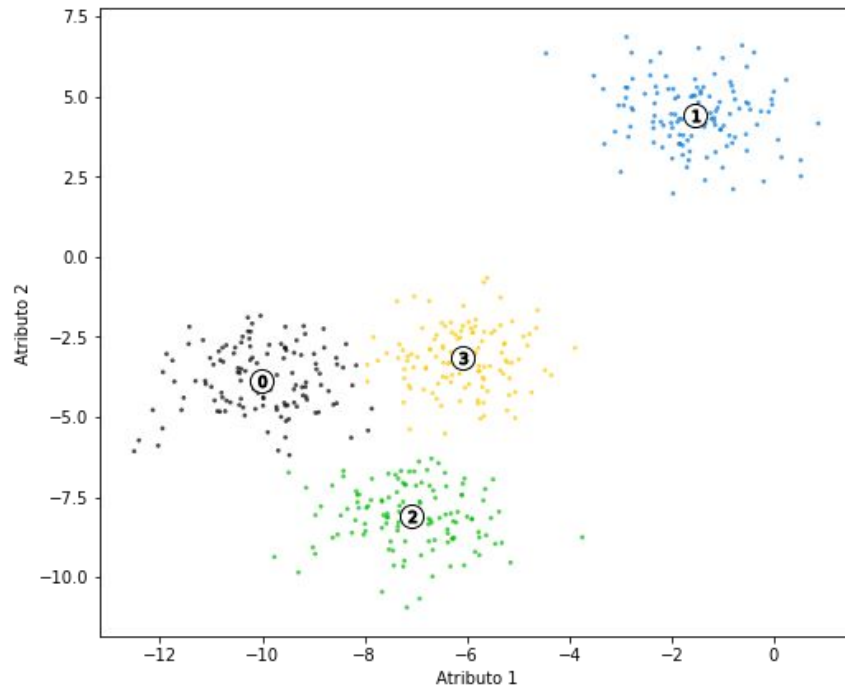
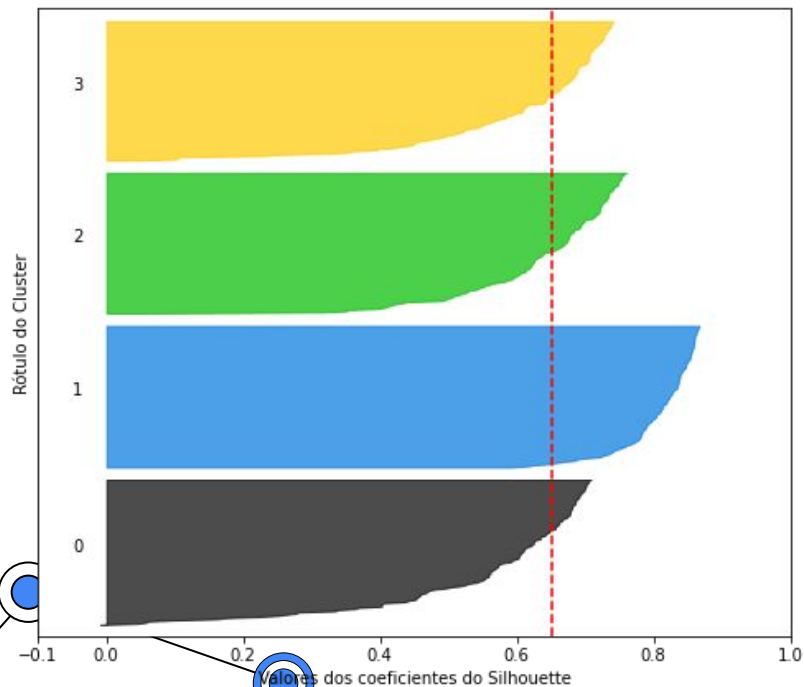


$k = 3$ (Silhouette avg : 0.58)



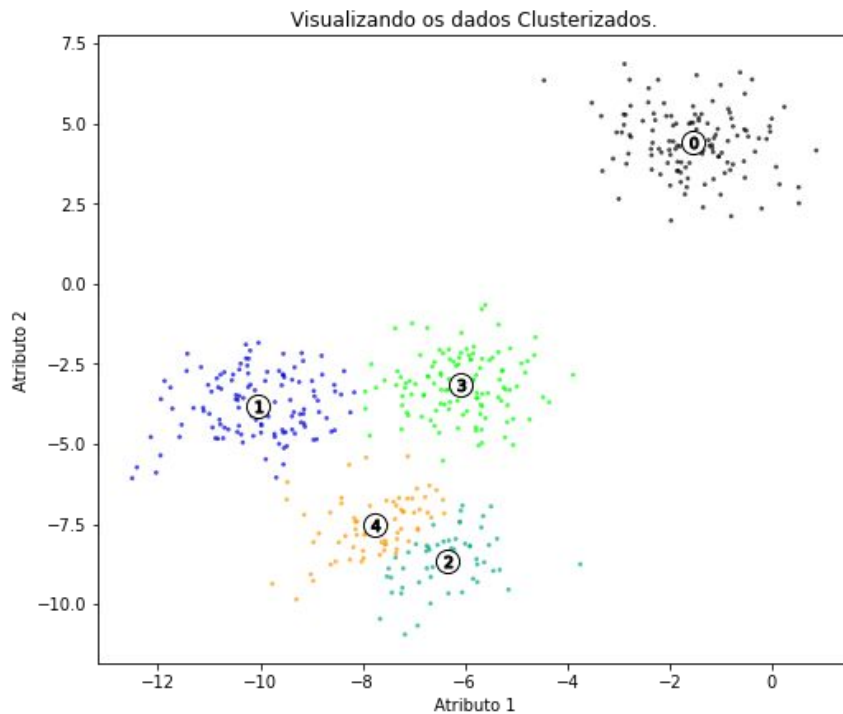
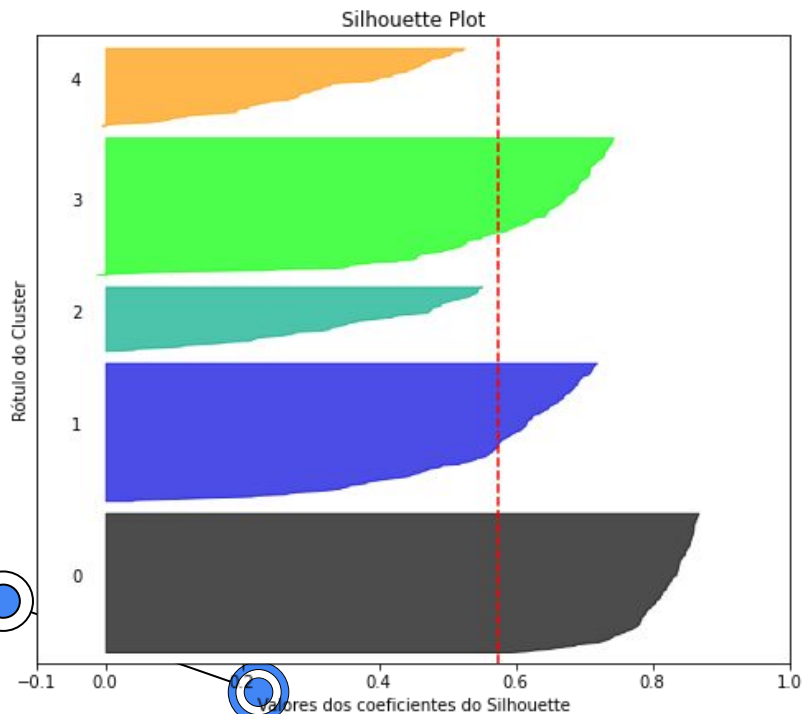
Silhouette Analysis

k = 4 (Silhouette avg : 0.65)



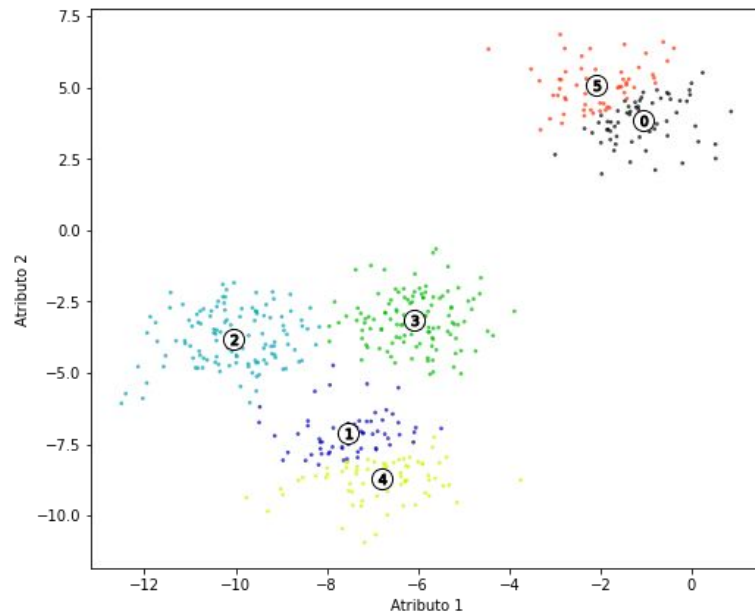
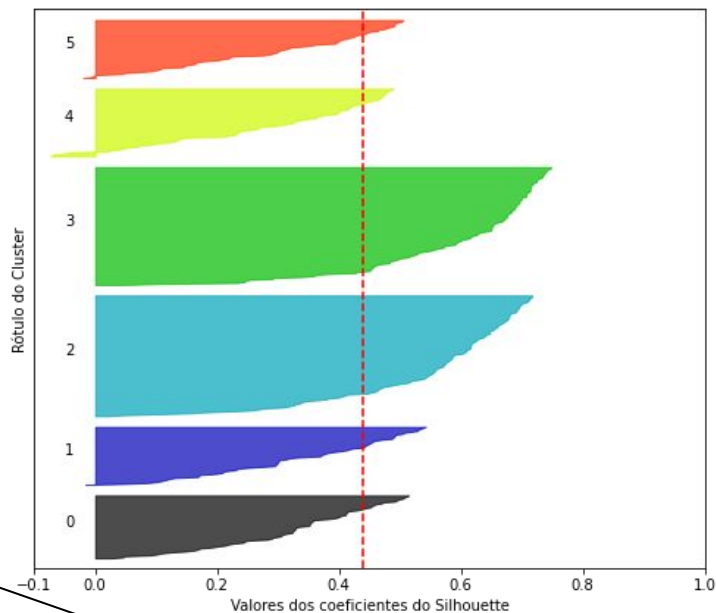
Silhouette Analysis

k = 5 (Silhouette avg : 0.57)



Silhouette Analysis

k = 6 (Silhouette avg : 0.43)



Challenges and Considerations in Clustering for MLOps

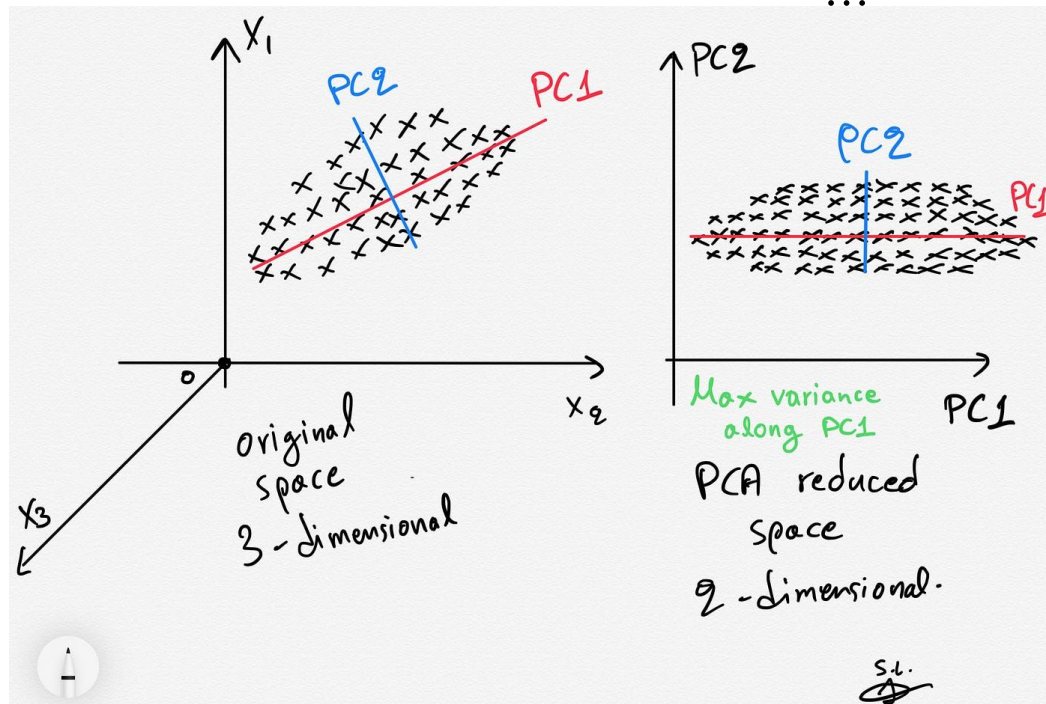
- **Scalability:** Many clustering algorithms can be computationally intensive, especially when dealing with large datasets. MLOps practitioners need to consider the **scalability of clustering algorithms** and ensure that they can handle the volume of data generated in production systems.
- **Interpretability:** Clustering results should be **interpretable to provide meaningful insights**. However, some clustering algorithms, such as deep learning-based methods, may produce complex or less interpretable results. Balancing the need for accuracy with interpretability is a key consideration in MLOps.
- **High-Dimensional Data:** Clustering high-dimensional data can be challenging due to the **curse of dimensionality**. High-dimensional data may lead to reduced cluster quality and increased computational complexity. Dimensionality reduction techniques or feature selection should be considered to mitigate these challenges.

Unsupervised ML, beyond clustering:

- **Dimensionality Reduction:**
 - Principal Component Analysis (PCA): Reduces the dimensionality of data while retaining as much variance as possible.
 - t-Distributed Stochastic Neighbor Embedding (t-SNE): Reduces dimensionality for visualization by preserving pairwise similarities between data points.
- **Anomaly Detection:**
 - Detects rare items or observations in a dataset that deviate significantly from the norm. Common methods include Isolation Forest, One-Class SVM, and Autoencoders.
- **Feature Learning:**
 - Automatically learns useful representations or features from the data. Techniques like autoencoders and deep belief networks are used for this purpose.
- **Sequential Pattern Mining:**
 - This task aims to discover frequently occurring sequences or patterns within a dataset. It is widely used in market basket analysis, web clickstream analysis, and recommendation systems. Popular algorithms for sequential pattern mining include AprioriAll and PrefixSpan.
- **Episode Mining:**
 - Episode mining extends sequential pattern mining to find more complex patterns that involve events happening concurrently or sequentially. It's often applied in areas such as healthcare for identifying concurrent medical events in patient records.

Principal Component Analysis ???

1. **Standardize the Data:** This step ensures that no single variable dominates the PCA process due to its scale.
2. **Compute the Covariance Matrix:** The covariance matrix describes how features in the dataset vary together
3. **Eigendecomposition of the Covariance Matrix:** The eigenvalues represent the variance along the principal components, while the eigenvectors represent the direction of these components.
4. **Sort Eigenvalues** and Select Principal Components
5. **Transform the Data**



Back to MLOps

